

2. Solve the following everyday problems.

- Sophie has a mobile phone plan. The plan costs \$60 per month. Sophie pays $\frac{3}{4}$ of the bill and her Mum pays the rest. How much does Sophie pay?
- Sophie makes $\frac{3}{2}$ of her phone calls between 9 am and noon. She makes $\frac{6}{1}$ of the calls between noon and 3 pm. What fraction of her calls is made between 9 am and 3 pm? What fraction is made between 3 pm and 9 am?
- Sophie talks to Ella on her mobile for $5\frac{4}{3}$ mins. She talks to Paul for $4\frac{1}{2}$ mins. How much longer did she talk to Ella than Paul? Sophie talks to Ella for twice as long the next day. How long was this phone call?
- Sophie is voting for 'The Y Factor' winner on her mobile phone. Each vote takes $\frac{1}{2}$ min to record. How many votes can Sophie make continuously in 10 minutes?

OLNA Practice Questions

- Last year, the number of people that went to the 'Groove in the South' music festival was 14 000. The number of people who went to the music festival this year was $\frac{5}{7}$ of last year's figure. How many people went to the festival this year?
A. 2000
B. 10 000
C. 9 500
D. 12 000
- What is the number exactly halfway between $2\frac{4}{3}$ and $3\frac{3}{4}$?

Topic 3

Using Mental Strategies to Solve Problems Involving Decimals

Mathematics Discussion

In Units 1 and 2 of the Foundations Mathematics course, we used mental strategies such as place value, partitioning and our knowledge of basic facts to solve problems involving whole numbers, money and the four operations. We used written jottings to perform calculations we could not completely store in our heads. These jottings included diagrams, number lines or number calculations. For problems involving 'difficult' numbers, we used a calculator.

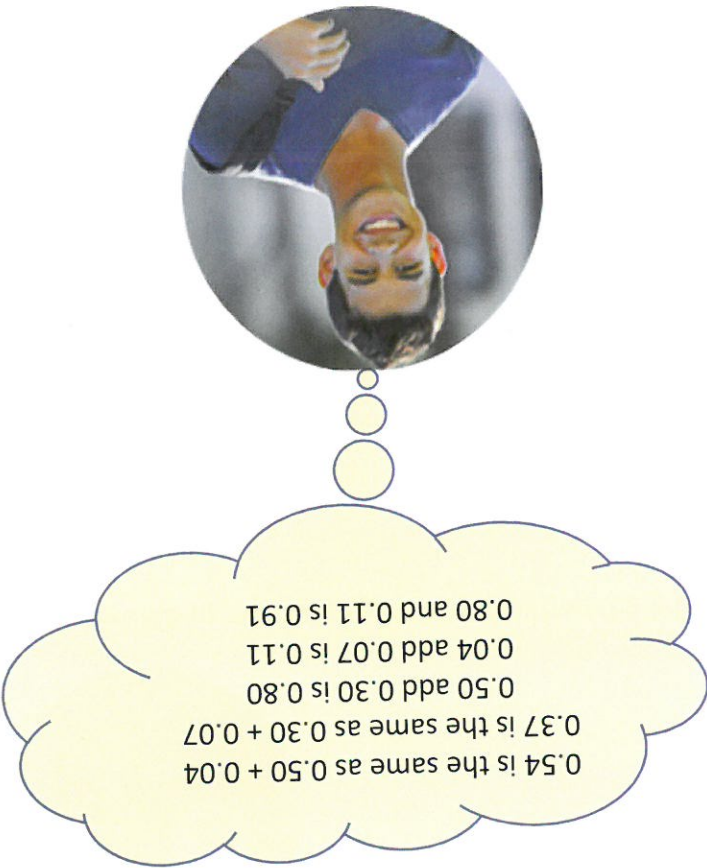
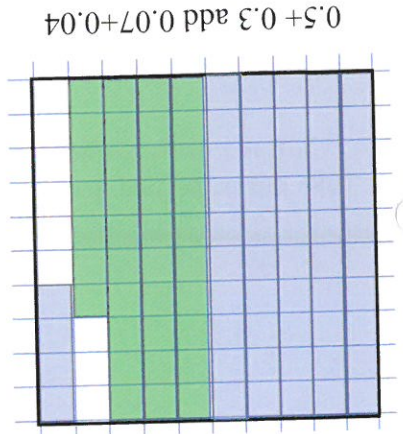
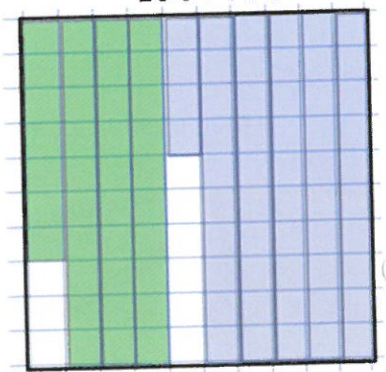
In this topic, we will work through exactly the same decision making, except we are now extending our work with whole numbers to include problems involving decimals.

Whole Class Activity 1

Think: What mental strategies can we use to solve addition problems involving decimals?

Jalen's Strategy for Adding Decimals

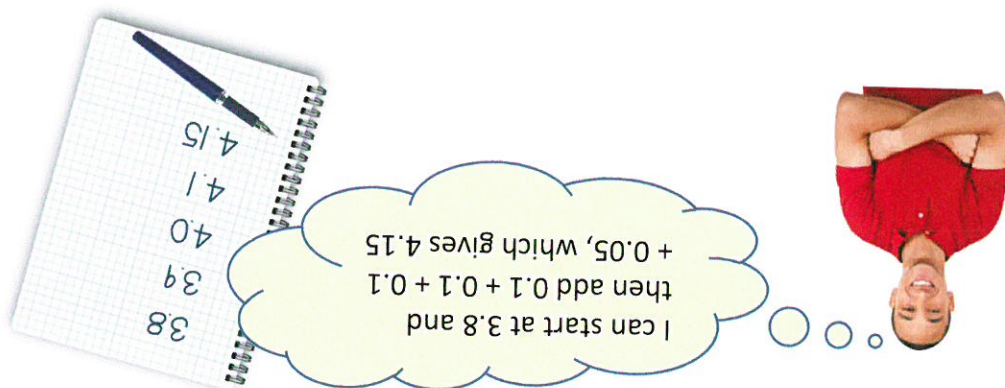
Jalen partitioned a whole square into 100 parts so he could use it to solve $0.54 + 0.37$. He used place value to partition the numbers. He coloured in blue, 5 tenths and 4 hundredths. He coloured in green, 3 tenths and 7 hundredths. He adds the tenths first and then the hundredths.



Jalen could see that the 5 tenths and 3 tenths equals 8 tenths. He could see that the 4 hundredths and 7 hundredths equals 11 hundredths, which is the same as 1 tenth and 1 hundredth. 8 tenths added to 1 tenth and 1 hundredth equals 9 tenths and 1 hundredth in total (i.e. 0.91)

1. Use Jalen's thinking and grid paper to solve the following problems.

- $1.2 + 0.3$
- $1.3 + 0.48$
- $2.4 + 1.7$
- $0.64 + 0.72$



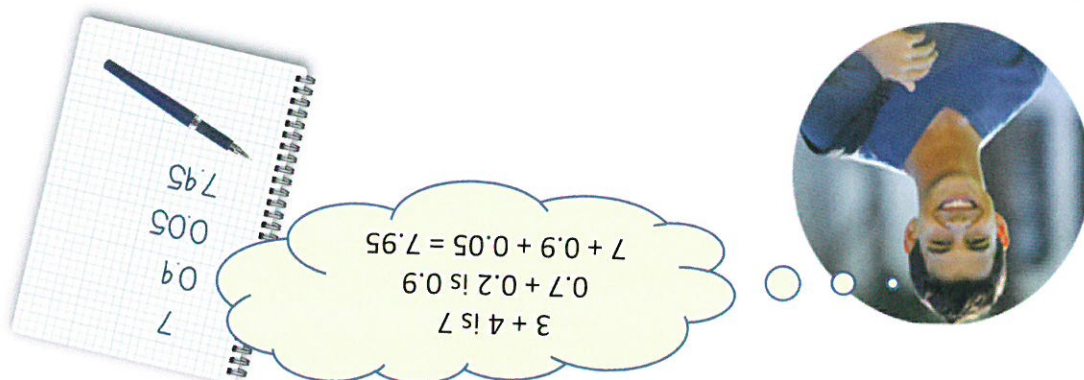
To solve a problem like $3.8 + 0.35$, Casey thought and wrote the following:

Casey realized that he could solve this decimal problem by counting forward, just as he did with adding whole numbers. He broke up 0.35 into tenths and hundredths. He then started at 3.8, added each of the tenths and then added the hundredths.

Casey and Paddo were solving $3.8 + 0.35$.

Casey and Paddo's Strategies for Adding Decimals

2. Use Jalen's method to mentally solve by visualizing the grid paper. Use jottings to help if necessary.
- a) $3.3 + 4.2$
 - b) $0.5 + 2.8 + 0.1$
 - c) $47.1 + 32.8$
 - d) $0.63 + 0.18$
 - e) $1.86 + 0.05 + 2$
 - f) $5.4 + 2.92$

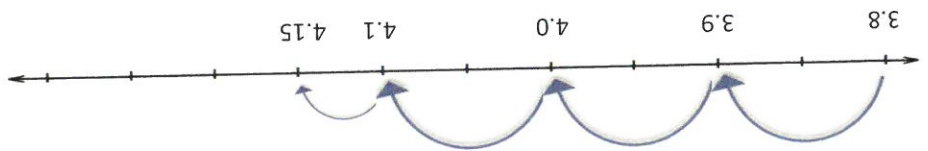


Jalen decided he could solve problems like these mentally, by visualizing the whole numbers, tenths and hundredths on grid paper.

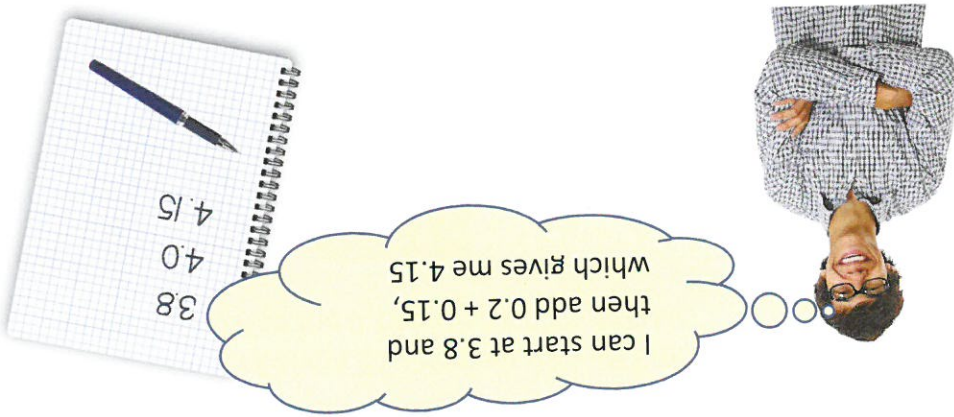
However, he found he sometimes needed to write down a few numbers to keep track of where he was up to.

To solve a problem like $3.7 + 4.25$, Jalen thought and wrote the following:

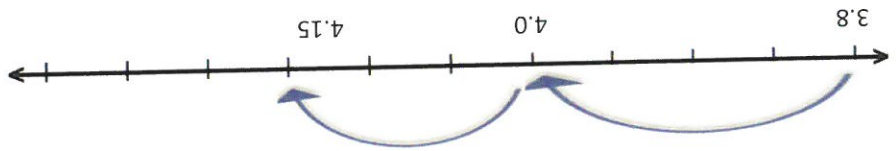
This can also be shown on a number line.



Paddo decided he could solve this problem just as he did whole numbers, by partitioning decimals and leaping along a number line. He started at 3.8. He split 0.35 into 0.2 and 0.15. He added 0.2 to 3.8 to get 4 and then finally added 0.15 to get 4.15



Paddo jotted down the following number line and made the leaps as shown:



3. Use Casey's thinking of counting forward or Paddo's partitioning and leaping along a number line to solve the following problems

a) $2.8 + 0.4$	
b) $3.7 + 1.5 + 0.2$	
c) $70.5 + 3.92$	
d) $45.8 + 2.6$	

c) $12.4 + 6.8$

b) $5.9 + 1.3 + 2$

a) $2.8 + 0.4$

6. Use Sarah's method of subtracting a decimal amount from one number and adding it to the other, to mentally solve the following problems. Use jottings or diagrams to help if necessary.

d) $8.99 + 1.32$

e) $1.77 + 1 + 0.9$

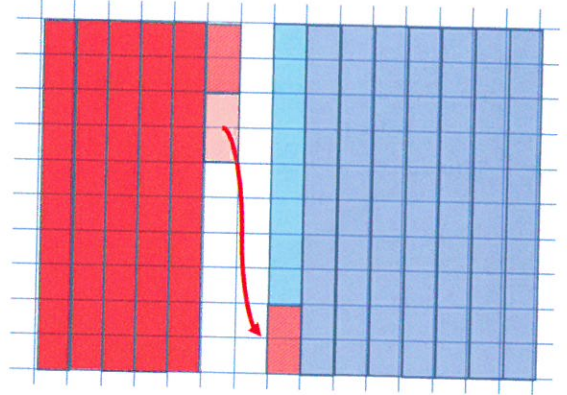
f) $2.31 + 1.9$

b) $0.9 + 1.7$

a) $1.9 + 0.4$

5. Use grid paper and Sarah's thinking to help solve the following problems

When Sarah calculated $2.7 + 0.45$, she jotted down $3 + 0.15$.
Can you see what Sarah did to these two numbers? Explain her thinking.



She first coloured in 0.78 and 0.54. She then took off 0.02 from the 0.54 and attached it to the 0.78 to make 0.8 add 0.52.

Sarah thought of $0.78 + 0.54$ as being $0.8 + 0.52 = 1.32$

Sarah's Strategy for Adding Decimals

c) $7.4 + 1 + 4.8$

b) $16.3 + 1.54$

a) $4.8 + 0.8$

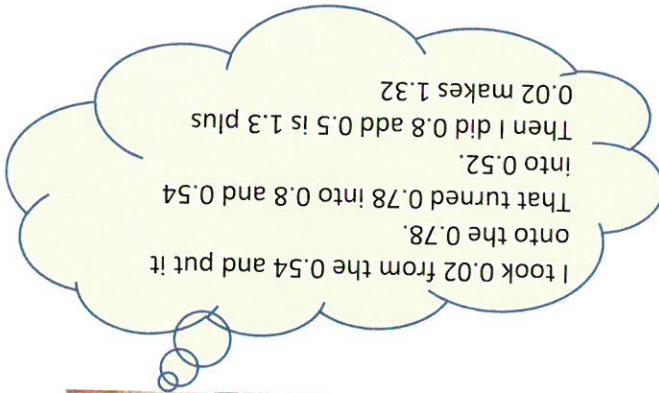
necessary.

4. Use Casey or Paddo's methods to mentally solve the following problems by visualizing counting forward or partitioning and leaping along a number line. Use jottings or diagrams to help if

d) $3.43 + 0.41$

e) $0.5 + 4.66$

f) $7 + 0.01 + 9.9$



Revision: The Best Strategy

Choose the most suitable strategy to solve the following problems. Try to do the problems mentally. If this is difficult, use jottings to help keep track

a) $0.53 + 0.32$	
b) $0.99 + 0.21$	e) $21.65 + 10.2$
c) $1.75 + 0.3 + 0.2$	f) $0.95 + 2.35$

Whole Class Activity 2

Think: What mental strategies can we use to solve subtraction problems involving decimals?

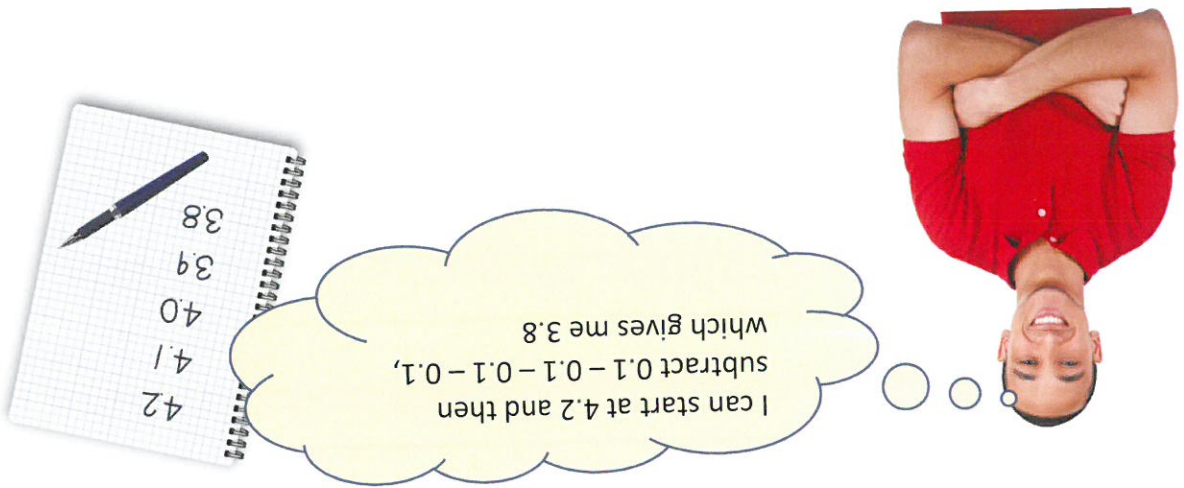
Casey and Paddo's Strategies for Subtracting Decimals

Casey and Paddo wondered whether they could use their number line thinking to do subtraction problems with decimals.

They were solving $4.2 - 0.4$.

Casey thought he could solve this decimal problem by counting backward, just as he did with subtracting whole numbers. He broke up 0.4 into tenths. He then started at 4.2 and subtracted backwards in tenths.

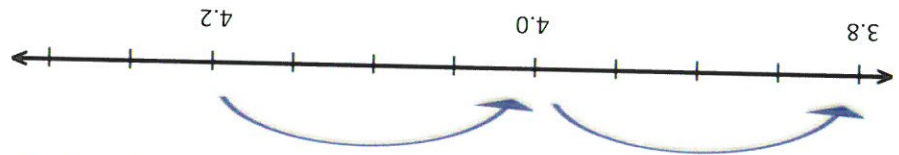
To solve a problem like $4.2 - 0.4$, Casey thought and wrote the following:



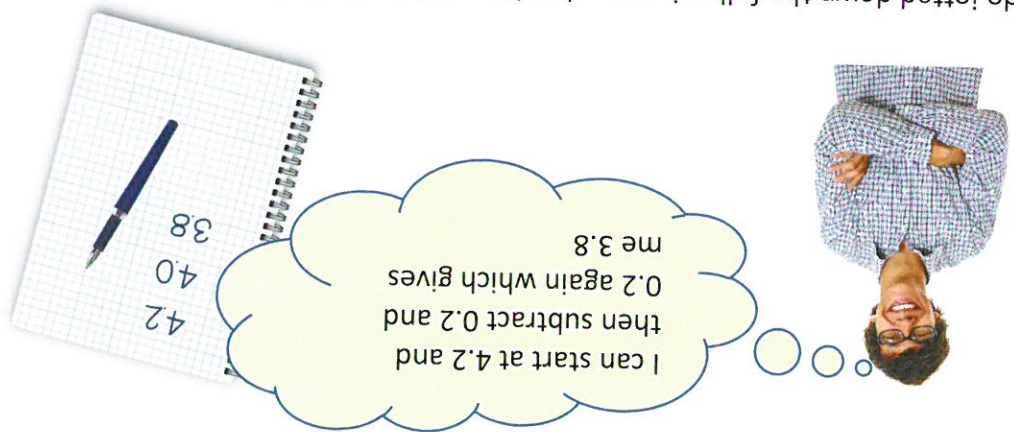
8. Use Casey or Paddo's methods to mentally solve the following problems by visualizing counting backwards or partitioning and leaping back along a number line. Use jottings or diagrams to help if necessary.

a) $4.3 - 0.5$	
b) $5.7 - 3.3$	
c) $0.7 - 0.51$	
d) $12.7 - 0.75$	

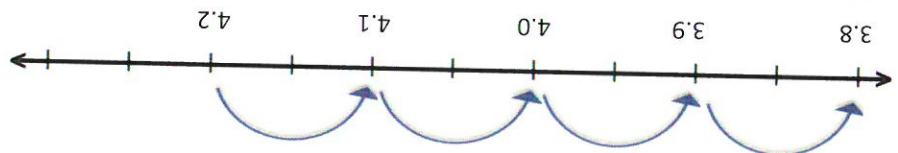
7. Use Casey's thinking of counting backward or Paddo's partitioning and leaping back along a number line to solve the following problems



Paddo jotted down the following number line and made the leaps as shown:



Paddo started at 4.2 on a number line. He partitioned 0.4 into 0.2 and 0.2 and then subtracted 0.2 from 4.2 to get 4 and then subtracted 0.2 again to get 3.8



This can be shown on the following number line.

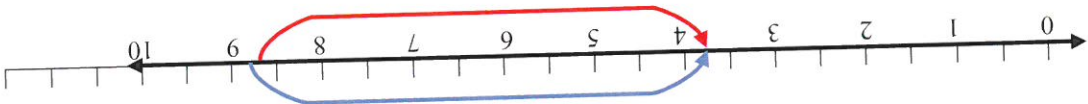
10. Use Lauren's method of adding or subtracting a decimal from both parts of the question to mentally solve the following problems. Use jottings or diagrams to help if necessary.
- a) $3.15 - 0.95$
d) $4.1 - 3.95$

a) $4.6 - 0.9$	
b) $8.2 - 5.8$	
c) $5 - 1.95$	
d) $21 - 1.01$	

9. Use Lauren's strategy to calculate the following on a number line. Write the new calculation under the existing one.

Adding or subtracting the same decimal to each number in a subtraction problem is a way of making it easier to calculate mentally.

Similarly, when Lauren calculated $9 - 1.01$, she jotted down $8.99 - 1$. Can you see what Lauren did to these two numbers? Explain her thinking.



To solve $8.7 - 4.9$, Lauren used a number line too. However, she added 0.1 to each of the numbers, changing $8.7 - 4.9$ into $8.8 - 5$. This works because the difference between 8.7 and 4.9 is the same as the difference between 8.8 and 5 . She started at 8.8 on the number line and jumped back 5 .

Lauren's Strategy for Subtracting Decimals

- a) $2.4 - 0.7$
b) $15.1 - 0.3$
c) $2.1 - 0.15$
d) $13.75 - 13.69$
e) $0.5 - 0.47 - 0.01$
f) $1.01 - 0.05$

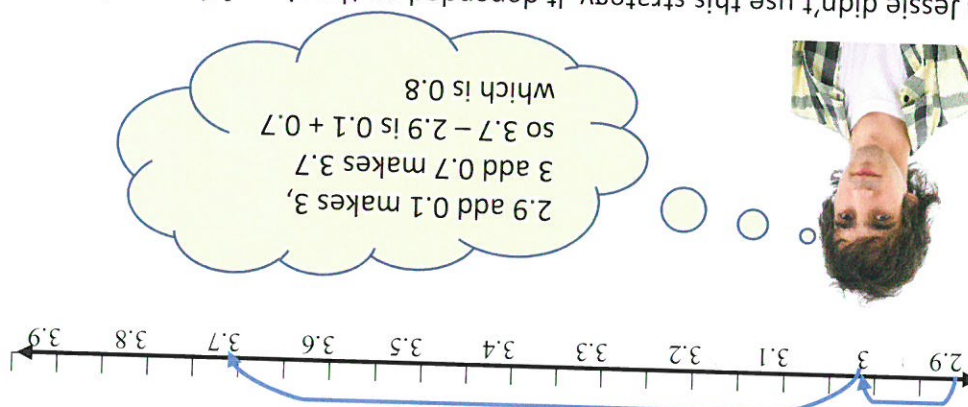
a) $2.7 - 2.4$		
b) $3.6 - 2.9$		
c) $1.15 - 0.9$		
d) $4 - 2.9$		

11. Use Jessie's strategy of placing the numbers in the Part-Part Whole diagram and counting forward to calculate the following:

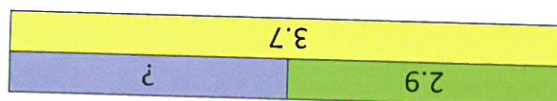


Why would it be easier to solve $3.7 - 2.9$ as an addition ($2.9 + ? = 3.7$) and $6.7 - 2.8$ as a subtraction?

Sometimes Jessie didn't use this strategy. It depended on the size of the numbers in the problem. If the numbers were $6.7 - 2.8$, Jessie would think of this as subtraction and solve it using either Paddo's or Lauren's method.



He then wrote the number sentence: $2.9 + ? = 3.7$
 Jessie then counted forward using jottings on a number line to solve the problem by addition



Jessie placed the numbers into the Part-Part Whole diagram as follows:

Jessie uses the Part-Part Whole model to help solve a variety of subtraction problems. When solving a problem such as $3.7 - 2.9$, Jessie used the model to change the problem from a subtraction into an addition.

Jessie's Strategy for Subtracting Decimals

- b) $3.6 - 0.9$
 c) $6 - 0.99$
 e) $3.05 - 2.98$
 f) $3.1 - 1.2$

12. Use Jessie's method of changing a subtraction into an addition to solve the following problems. Use jottings or diagrams to help if necessary.

- a) $5.1 - 4.8$
- b) $2.3 - 1.7$
- c) $11.7 - 10.9$
- d) $12.1 - 11.94$
- e) $6.1 - 5.95$
- f) $4.05 - 3.96$

Revision: The Best Strategy

Choose the most suitable strategy to solve the following problems mentally. Use jottings to help keep track.

a) $3.4 - 1.9$	d) $34.7 - 23.3$
b) $15.15 - 14.85$	e) $8.3 - 7.8$
c) $1.2 - 0.3 - 0.1$	f) $6 - 1.8 - 0.2$

Practice Exercise 1

1. PUZZLE:

'What did Cinderella wear at the beach?'
Complete these questions mentally to decode the answer to this joke.

- S: $5.1 - 4.8$
- i: $3.2 + 1.6 + 4$
- I: $6.2 - 0.9$
- L: $0.54 + 1.6$
- G: $0.95 + 1.35$
- P: $8 - 0.8 - 0.8$
- R: $1.99 + 2.06$
- E: $17.1 - 0.35$
- A: $5.68 + 0.4 - 0.01$
- S: $23.4 + 2.85$

5.3	2.3
6.4	2.14
13.8	6.07
16.75	0.3
4.05	26.25
26.25	1.25
8.8	2.14

2. For the following problems:
- Write the number sentence to reflect the situation and the number sentence to calculate the answer (these may be the same)
 - Show the strategy chosen for solving the problem
 - Find the answer
 - Think about your answer. Does it make sense?

a) Austria recycles 49.7% of its waste. Switzerland recycles 52.1%. How much more does Switzerland recycle than Austria?

Number Sentence(s)	Strategy	Answer	Does it make sense?

b) The amount of recycling by each Australian increased from 0.08 tonnes per person in 1997 to 0.75 tonnes in 2003. By what tonnage had the recycling increased in this period?

Number Sentence(s)	Strategy	Answer	Does it make sense?

c) Taiwan's recycling industry was worth 0.8 billion dollars in 2005. In a decade, it rose 1.7 billion dollars. How much was it worth in 2015?

Number Sentence(s)	Strategy	Answer	Does it make sense?

d) Between 1997 and 2011, Taiwan slashed daily household waste per person from 1.15 kg to 0.45 kg. How much had the daily household waste decreased?

Number Sentence(s)	Strategy	Answer	Does it make sense?

e) In 1997, the yearly waste generated by Australians was 1.23 tonnes. This grew by 0.4 tonnes over the next 6 years. How much waste was each Australian generating in 2003?

Number Sentence(s)	Strategy	Answer	Does it make sense?

f) Switzerland recycles 52.1% of its waste. The U.S. recycles 9.5% less. What percentage does the U.S. recycle?

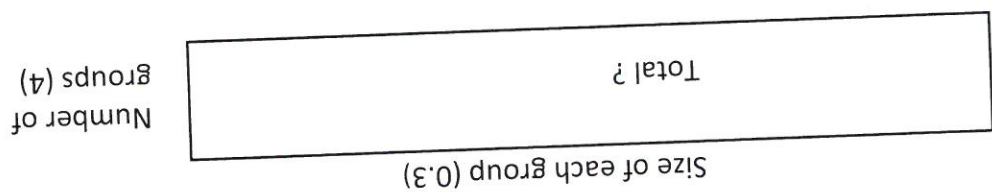
Number Sentence(s)	Strategy	Answer	Does it make sense?

Whole Class Activity 3

Think: What mental strategies can we use to solve multiplication problems involving decimals?

Dylan's Array Strategy for Multiplying Decimals

Dylan uses an array diagram to solve some multiplication of decimal problems. For example, to solve 4×0.3 he placed the numbers in the array diagram.



He started at 0.3 and counted forwards by 0.3, four times. He was thinking of the problem as $0.3 + 0.3 + 0.3 + 0.3$ which equals 1.2.

1. Use your knowledge of arrays and repeated equal quantities to solve the following problems. Use jottings or diagrams to help if necessary.

- a) 0.2×4
- b) 3×0.12
- c) 7×0.2
- d) 0.5×9
- e) 1.2×6
- f) 0.27×3

THOUSANDS			ONES			DECIMALS	
Hundreds	Tens	Ones	Hundreds	Tens	Ones	Tenths	Hundredths
			2	4	5		
				2	4	•	5
					2	•	4
							5

Using Patterning and Place Value for $\times 10$

We can use the Place Value Chart and Patterning to solve $\times 10$ decimal problems mentally. For example, to solve 2.45×10 we can place 2.45 in the chart with the 2 in the Ones column, the 4 in the Tenths column and the 5 in the Hundredths column.

Multiplying by 10 makes the number 10 times larger. Every number moves one place to the left on the chart. Therefore, $2.45 \times 10 = 24.5$.

If we were to multiply 2.45 by 100, every number in 2.45 would be 100 times larger and it would move two places to the left on the chart. (i.e. 245)

2. Use Tom's strategy of doubling to calculate the following. Use jottings or diagrams to help if necessary.
- a) 4.5×2
 - b) 4×3.12
 - c) 8×1.25
 - d) 3.34×2
 - e) 4×10.5
 - f) 2×11.6

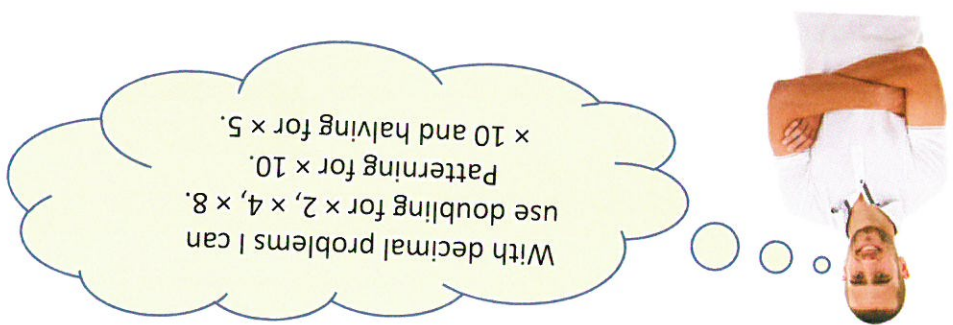
Using doubling for $\times 2$, $\times 4$ and $\times 8$

We can extend the doubling we used for whole numbers to mentally solve decimal problems involving $\times 2$, $\times 4 \times 8$. For example,

To solve 2.25×2 we can double 2.25 to get 4.5.

To solve 2.25×4 we can double 4.5 to get 9

To solve 2.25×8 we can double 9 to get 18



Tom's Strategy for Multiplying Decimals

Tom knows he can extend the strategies he developed in Multiplication of Basic Facts (Unit 2) to solve multiplication problems involving decimals.

Katie's Strategy for Multiplying Decimals

Using $\times 10$ and halving for $\times 5$

[illegible]

- a) 1.5×4
- b) 2.5×18
- c) 6×11.5
- d) 3.5×12
- e) 0.8×1.5
- f) 0.5×0.7

6. Use Penny's strategy of doubling one number and halving the other to solve the following problems. Use jottings to help if necessary.



What types of multiplication of decimal problems are best suited to Penny's method?

Penny said 'I can solve 0.6×0.5 by halving 0.6 and doubling 0.5 but I wouldn't choose that method to solve 0.3×0.7 '



I can think of 12 as 2×6 .
So 2.5×12 is the same as $2.5 \times (2 \times 6)$
I can rearrange the brackets.
 $(2.5 \times 2) \times 6 = 5 \times 6$
That's how doubling and halving works.

In some multiplication of decimal problems, Penny can double one of the numbers and halve the other to make the mental calculation easier. For example,
To solve 2.5×12 , she can double 2.5 and halve 12 i.e. $2.5 \times 12 = 5 \times 6$

Penny's Strategy for Multiplying Decimals

5. Use Katie's strategy of partitioning numbers to solve these problems. Use jottings to help.
- a) 1.1×3
 - b) 2.3×4
 - c) 6×0.12
 - d) 7×4.2
 - e) 9×1.5
 - f) 10.4×6

a) 0.2×6	
b) 0.17×3	
c) 4×2.6	
d) 5×16.6	
e) 0.15×8	
f) 100×6.8	

Choose the most suitable strategy to solve the following problems. Try to do the problems mentally. If this is difficult, use jottings to help keep track

Revision: The Best Strategy

NOTE: Most people tend to use written jottings to help keep track of multiplication calculations that cannot be completely stored in their heads. For problems involving 'difficult' numbers a calculator should be used.

Justify your answers and circled problems with a partner. Did you agree? Why? Why not?

- 0.5×12.6
 - 31.7×100
 - 16×0.25
 - 0.1×17.2
 - 8×1.5
- Calculate the answers to all these problems. Circle the problems where it was easier to think of a decimal as a fraction.

Sometimes, it is easier to solve a multiplying decimals problem by thinking of the fraction equivalent of a decimal within the problem. For example, it is easier to solve:

- 0.5×4 by thinking of it as $\frac{1}{2}$ of 4 which is 2.
- 0.25×6 by thinking of it as $\frac{1}{4}$ of 6 i.e. halving 6 and halving it again which is 1.5

Reflection and Discussion

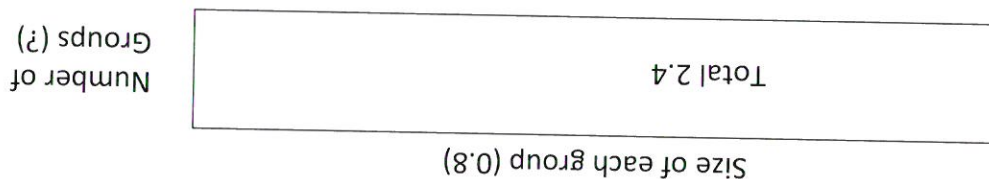
Think: What mental strategies can we use to solve division problems involving decimals?

Dylan's Array Strategy for Dividing Decimals

Dylan used the array model to help solve division problems.

a) To solve $2.4 \div 0.8$, he decided that it made sense to think 'how many groups of 0.8 in 2.4' rather than '2.4 shared between 0.8'.

He placed the numbers into an array diagram. He did not yet know the number of groups.

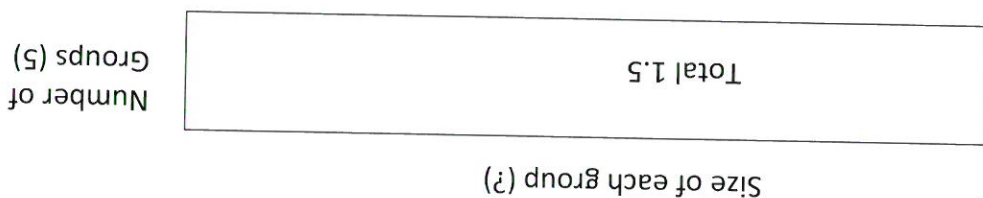


He thought of the problem as $? \times 0.8 = 2.4$

To work it out he started at 0.8 and counted forward by 0.8 until he got to 2.4. That is, $0.8 + 0.8 + 0.8 = 2.4$. He worked out there was 3 groups of 0.8 in 2.4.

b) To solve $1.5 \div 5$, Dylan decided it was easier to think of it as 'sharing or partitioning 1.5 into 5 equal parts'

He represented the problem as an array



He thought of this problem as $5 \times ? = 1.5$.

He knew $5 \times 3 = 15$. That was ten times bigger than he needed. He thought that 1.5 shared into 5 equal parts would be 0.3. He tested that idea using multiplication and decided he was correct.

1. Use your knowledge of arrays and repeated equal quantities to solve the following problems. Use jottings or diagrams to help if necessary.

a) $2.5 \div 5$

b) $12.3 \div 3$

c) $5.5 \div 1.1$

f) $15 \div 0.5$

e) $0.21 \div 7$

d) $6 \div 0.25$

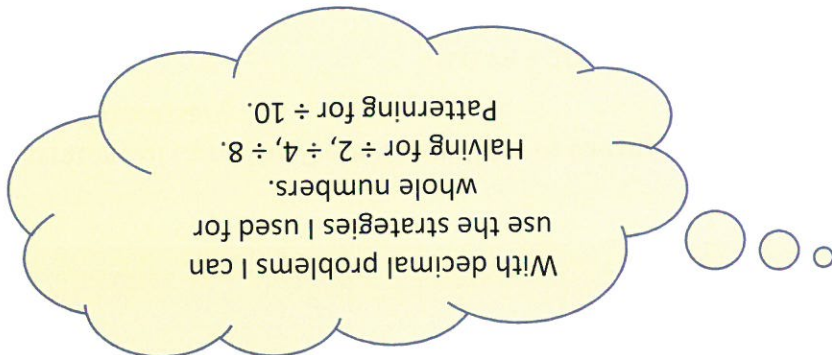
Which problems were easier to work out as sharing problems?

Which problems were easier to work out as grouping problems, i.e., 'how many groups of'?



Tom's Strategy for Dividing Decimals

Tom knows he can extend the strategies developed in Division of Basic Facts (Unit 2) to solve decimal problems.



Using halving for $\div 2$, $\div 4$ and $\div 8$

We can extend the halving we used for whole numbers to solve decimal $\div 2$, $\div 4$, $\div 8$ problems mentally. For example,

To solve $14.4 \div 2$ we can halve 14.4 to get 7.2.

To solve $14.4 \div 4$ we can halve 14.4 to get 7.2 and halve it again to get 3.6.

To solve $14.4 \div 8$ we can halve 14.4 to get 7.2, halve it again to get 3.6 and halve it again to get 1.8.

2. Use Tom's strategy of halving to calculate the following. Use jottings or diagrams to help if necessary.

a) $3.2 \div 2$

b) $4.44 \div 4$

c) $0.48 \div 8$

d) $36.5 \div 2$

e) $0.6 \div 4$

f) $3 \div 4$

Choose the most suitable strategy to solve the following problems. Try to do the problems mentally. If this is difficult, use jottings to help keep track

NOTE: Most people use written jottings to help keep track of division calculations that cannot be completely stored in their heads. For problems involving 'difficult' numbers, a calculator should be used.

3. Use Tom's strategy of using the Place Value Chart to calculate the following. Use the spaces in the chart to help if necessary.

We can use the Place Value Chart and patterning to solve $\div 10$ decimal problems mentally. For example, to solve $12.7 \div 10$ we can place 12.7 in the chart with the 1 in the Tens column, 2 in the Ones column and 7 in the Tenths column.

Dividing by 10 makes the number 10 times smaller. Every digit in 12.7 moves one place to the right. Therefore, $12.7 \div 10 = 1.27$

If we were to divide 12.7 by 100, every digit in 12.7 would be 100 times smaller and it would move two places to the right. Therefore, $12.7 \div 100 = 0.127$.

Using Patterning and Place Value for $\div 10$

Practice Exercises 2

1. PUZZLE:

'What's Green and Stands in a Corner?'

Complete these questions to decode the answer to this joke.

G: $0.65 \div 10$ R: 2.14×7 F: $15 \div 2.5$
T: $0.5 \times 3 \times 2$ G: $3 \div 0.25$ U: $0.2 \times 10 \times 10$
A: 0.65×4 H: 0.3×9 N: $3.39 \div 3$
A: $2 \div 8$ O: $24.4 \div 4$ Y: $14.2 \times 10 \div 2$

0.25	1.13	2.6	20	12	2.7
3	71	6	14.98	6.1	0.065

2. For the following problems:

- Write the number sentence to reflect the situation.
- Write the number sentence to calculate the answer (this may be the same as above.
- Show the strategy chosen for solving the problem.
- Find the answer.
- Think about your answer. Does it make sense?

a) Americans use an average of 300.6 L of water per day per person. How much is used per person over 5 days?

Number Sentence(s)	Strategy	Answer	Does it make sense?

b) A washing machine uses 39.6 Litres of water. A more water-wise machine uses 3 times less. How much water does the water wise machine use?

Number Sentence(s)	Strategy	Answer	Does it make sense?

PROBLEM		ANSWER	PROBLEM	ANSWER
a) 7.5×4		0.85	g) $1.4 + 27.8$	0.6
b) $4.2 + 5 - 0.1$		40	h) $4.8 \div 8$	9.47
c) $10 \div 0.25$		6	i) $4.3 - 0.5$	0.7
d) $7.8 - 6.95$		70	j) $288.2 \div 10$	29.2

1. Calculate the answers to the following problems mentally. Use jottings and diagrams to help. Draw a line to connect the question with the correct answer.

Reflection on Learning

Number Sentence(s)	Strategy	Answer	Does it make sense?

f) A low water flow AAA rated showerhead uses 7.5 litres of water every minute. How many litres will an 8 minute shower use?

Number Sentence(s)	Strategy	Answer	Does it make sense?

e) An average toilet uses 5.5 litres of water in every flush. How many litres of water will it use in 12 flushes?

Number Sentence(s)	Strategy	Answer	Does it make sense?

d) A dishwasher uses 4.3 litres of water in 2 washes. How much water is used per wash?

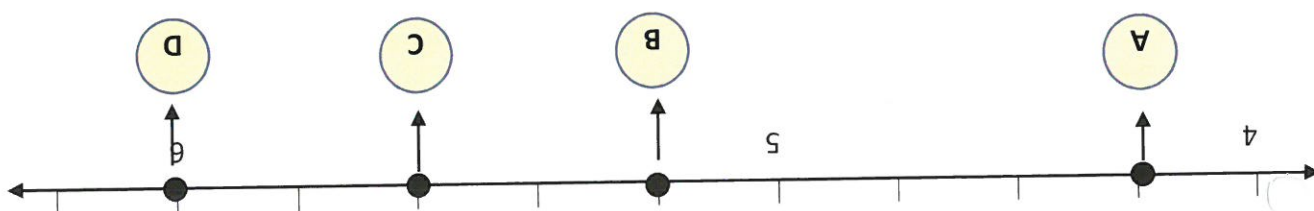
Number Sentence(s)	Strategy	Answer	Does it make sense?

c) In the U.S, baths account for 2.7% of household water usage. Showers account for 4 times more. What percentage of the household water usage do showers use?

million

3. The population of Australia in 1950 was 8.27 million. The population of Australia in 2015 was 23.9 million. What is the difference between these two populations?

2. Melba divided 72.9 by a number. She got 0.729 as the answer. What number did Melba divide by?



Which dot on the number line best represents this amount?

1. $4.7 + 0.5 = ?$

OLNA Practice Questions

3. Solve the following problems
- Australia's average temperature has increased by 0.9° since 1950. If the average temperature was 21.7° in 1950, what is it today?
 - Australia accounts for 1.48% of global greenhouse gas emissions. This is 4 times higher than the average greenhouse gas emissions of all countries across the world. What is the world average?
 - In 1990 Australia's greenhouse gas emissions were 547.7 million tonnes. In 2007 they totalled 597.2 million tonnes. How much have they increased in this time?
 - Biogas accounts for 0.3% of Australia's electricity generation. Hydroelectricity accounts for 7 times this amount. What percentage of electricity generation is hydro?

2. a) Circle in **blue**, any problems where your strategy involved thinking of a number line.
 b) Circle in **red**, any problems where your strategy involved doubling, halving or both.
 c) Circle in **green**, any problems where you thought of the Place Value Chart and patterning.
 d) Circle in **black**, any problems where it was easier to calculate using the fractional equivalent of the decimal.
- Check with a partner. Did you circle with the same colours? Justify your thinking.

e) $1000 \times 0.7 \div 10$	30	k) $8.97 + 0.5$	3.8
f) 24×0.25	9.1	l) $4.4 - 3.7$	28.82

Topic 4

Using a Calculator or Spreadsheet to Solve Everyday Problems

Mathematics Discussion

When solving everyday problems involving fractions and decimals, we decide:

- whether an approximate or exact solution is required
- whether to solve the problem mentally or with a digital resource such as a calculator, phone or spreadsheet
- the number sentences that reflect the problem and help us calculate the answer
- the strategy for finding the answer
- whether the answer is reasonable

In this section, we will be focussing on choosing between mental, calculator or spreadsheet strategies to solve problems involving fractions and decimals.

We tend to use mental strategies for one step or simple multistep problems involving easily visualized fractions and simple decimals. Examples of such problems include those with number sentences like $\frac{4}{3} + 1\frac{1}{2} - \frac{1}{4}$ and 1.2×3 .

We tend to use calculators when the numbers are more difficult, such as $1\frac{7}{5} + 10\frac{1}{3}$ and $3.72 \div 0.5 \times 1.23$ but the problem is still limited to either one or a few steps. We can solve fraction problems on a calculator with careful use of the division ($\div$) button and brackets on a basic calculator, or by using the fraction 'A/b/c' button on a scientific calculator.

We tend to use a spreadsheet when we need to calculate a lot of data or data that sometimes needs to be changed or added to. When entering fractions into a spreadsheet formula we use, for example $3/7$ for $\frac{3}{7}$ and $2\ 3/7$ for $2\frac{3}{7}$.